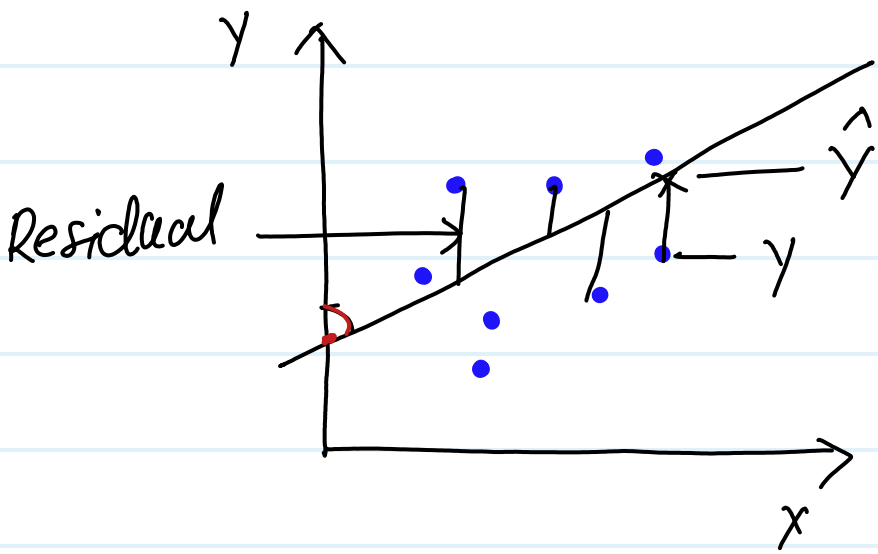
LR.



$$Y = mx + c$$

simple linear eqn

$$Y = c + mx$$

multi linear eqn

$$Y = c + m_1x_1 + m_2x_2 + \cdots + m_nx_n$$

$$h_{(\theta)} = \theta_0 + \theta_1 x_1$$

Loss $= y - \hat{y}$

Cost function $= \frac{1}{2m} \sum_{i=1}^{m} \left(h_{(\theta)} x_i - y_i \right)^2$

OR

$$= \frac{1}{2m} \sum_{i=1}^{m} \left(y_i - \hat{y}_i \right)^2$$

Repeat conversion theorem

$\theta_0 \quad \theta_1$

$$\theta_{0\,new} = \theta_{0\,old} - \alpha \frac{\partial L}{\partial \theta_{0\,old}}$$

$$\theta_{1\,new} = \theta_{1\,old} - \alpha \frac{\partial L}{\partial \theta_{1\,old}}$$

$$h_\theta(x) = \theta_0 + \theta_1 x_1$$

$\hat{y}$

50 $\qquad y - \hat{y}$

44

Evaluation metrics

1) MAE

$$= \frac{\sum |y - \hat{y}|}{n}$$

2) MSE $= \frac{\sum (y - \hat{y})^2}{n}$

3) RMSE $= \sqrt{MSE}$

4) R^2 score $= 1 - \frac{SS_{reel}}{SS_{Tot}}$

5) Adj R^2 score $= \frac{1 - (1 - R^2)(N-1)}{N-P-1}$

$$y = b_0 + b_1 x$$

$$\boxed{y = \underbrace{7.5}_{\theta_0} + \underbrace{41.5}_{\theta_1} x}$$

$$y - \hat{y}$$

$$50 - 48 = 2$$

$$\theta_0 = 7.5$$

$$\theta_1 = 41.5$$

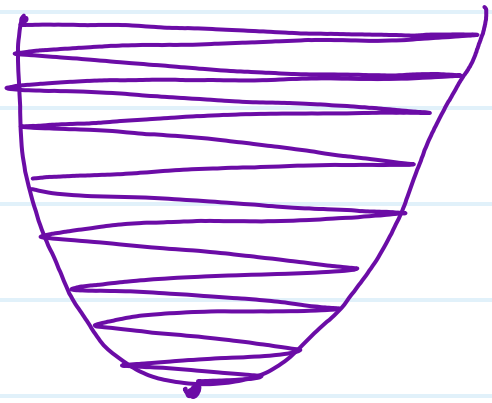
$$\theta_{0new} = \theta_{0old} - \alpha \frac{\partial L}{\partial \theta_{0old}}$$

$$= 7.5 - 0.5 \left(\frac{\partial 2}{\partial 7.5} \right)$$

$$\theta_{0new} = 6.2$$

$$\theta_{1new} = 41.5 - 0.5 \left(\frac{\partial 2}{\partial 41.5} \right)$$

$$\theta_{1new} = 41$$



$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6 \; x_7 \; Y$

No multicolinearity

VIF (variance inflation factor)

$$\Rightarrow \frac{1}{1+R^2}$$

x_1 – 10 ✗

x_2 – 2

x_3 – 4

x_4 – 7 ✗

x_5 – 1

x_6 – 2

x_7 – 3

$x_i > 5$

0
1
2
3
4
5

overfitting

score

train = 96 ✓

test = 60 ✓

Trade-off of Bias and variance

low Bias
high variance

underfitting

train = 60

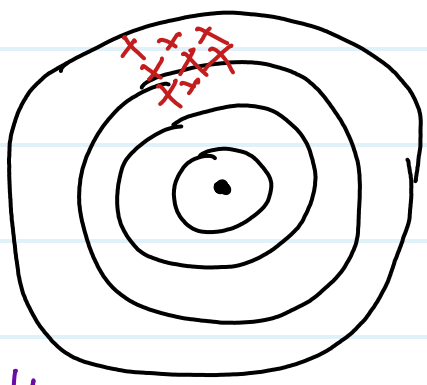
test = 50 / 85

high bias
low variance

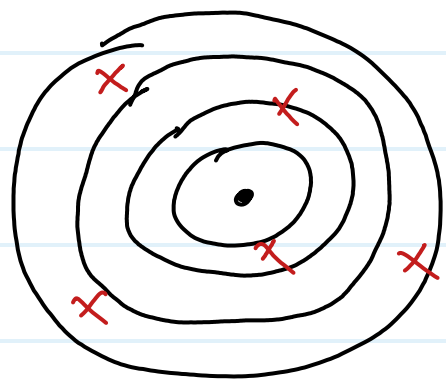
Best fit condition

train – 70
test – 68 / 72

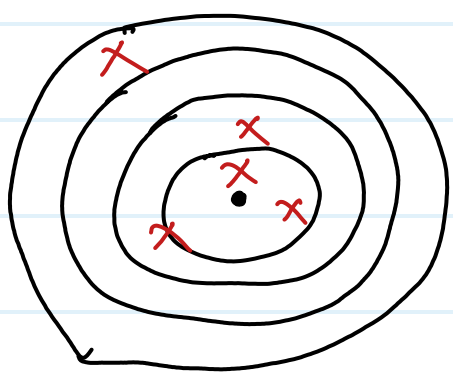
low bias
low variance



underfitting



overfitting



Best

Regularization / Generalization

$$y = c + mx$$

penalty term, constant = λ
(0.1, 0.5, 1, 0.01, 10)

Why Regularization?

→ coefficient become large.

x_1 size x_2 Room y Price

$$\text{price} = 5 + 20(\text{size}) + 30(\text{Rooms})$$

L_2 Regularization (Ridge)

$$\text{Loss} = \sum (y - \hat{y})^2 + \lambda \sum b^2$$

coeff $b_1 = 10$

$b_2 = 8$

penalty $\rightarrow$ $10^2 + 8^2$

$100 + 64$

164

$= \sum (y - \hat{y})^2 + \lambda (164)$

$\Rightarrow \sum (y - \hat{y})^2 + (0.1 \times 164)$

To prevent model from overfitting

L_1 Regularization (Lasso)

$$\text{Loss} = \sum (y - \hat{y})^2 + \lambda \sum |b|$$

$b_1 = 10$

$b_2 = 8$

$= 10 + 8$

$= 18$

$= \lambda \times 18$

It is used to select feature or reduce the feature

Elasticnet Regularization

$$Loss = \sum (y - \hat{y})^2 + \underbrace{\lambda b^2}_{L_2} + \underbrace{\lambda |b|}_{L_1}$$